

RD905UW / MS9 Setup Guide

UHF RFID Reader — Connection & Output Protocol Reference

Compiled from the MS9 documentation package (product sheets, configuration guides, SDK manual, and support notes).

1. What This Actually Is

RD905UW is a member of the RD905M mid-range UHF integrated reader/writer series, built on the “UHFRReader18” reference design and protocol — confirmed in the adapted-products list (适配产品说明.doc): “型号: RD905UW (应答模式有效)”. All software, DLLs, and protocol docs in the MS9 package (UHFRReader18 Demo Software, UHFRReader18.dll, SDKs) apply to it. Full spec sheet: 4.产品说明/RD905M 超高频远距离系列说明.doc.

- Size: 235 x 235 x 57 mm, approx. 900 g, 8 dBi antenna, ≥ 5 m typical read range
- Power: single +9V DC (operating range 8–12V), approx. 350–650 mA
- Frequency: FCC 902–928 MHz / CE 865–868 MHz (region-configurable)
- Standard interfaces: Wiegand + RS232 + RS485
- Optional: RJ45 (TCP/IP) / WiFi

2. Physical Connection

- Power the unit with the supplied +9V adapter.
- Wire the reader to your host via one of: RS232/RS485 direct to PC (or a USB-to-serial adapter); TCP/Ethernet (RJ45 variant) via LAN cable or straight to a PC NIC; or Wiegand 26/34 straight into a controller/access panel.
- Wire colors are NOT standardized across batches — trust the label printed on the back of your specific unit, not a generic pinout diagram.
- USB version: install the CP210x VCP driver first (folder: 8.USB 驱动/CP210x_VCP_Windows) — the reader then appears as a virtual COM port.
- Mounting bracket/pole diagram: 4.产品说明/安装示意图/RD905 安装参考图.png (bracket bolts to the back of the unit, then clamps to a pole up to 5 cm diameter).

3. Software Setup

Use UHFRReader18 Demo Software v2.6 (1.参数设置/参数设置软件 V2.6.exe).

Serial (RS232/RS485)

- Port: AUTO (or select the COM port)
- Baud: 57600 bps (default) — 8 data bits, 1 start bit, 1 stop bit, no parity
- Reader address: FF (broadcast — works regardless of the unit's actual address)
- Click “打开端口” (Open Port), then “获得读写器信息” (Get Reader Info) to confirm the connection

TCP / Ethernet

- Direct-connect a cable PC to reader, and set your PC's IP to the same subnet as the reader's default IP (approx. 192.168.1.x range)
- Disable Windows Firewall / antivirus — it blocks the discovery broadcast; if search still fails, connect PC to reader directly instead of through a switch
- Open the “TCP/IP 配置” tab in the demo software, click Search — default port 6000
- To push data to a server: set the reader as TCP client, target IP = your server, target port 6000 or higher; the reader auto-reconnects on the next tag read if an idle timeout drops the link

Important RD905UW-specific note (from 5.写EPC号/写EPC号软件/说明.txt): RD905UW is permanently fixed to “Answer / Response Mode” — it cannot be switched into free-running Scan Mode like other family members. The host must send a command (e.g. Inventory) and the reader replies once per request.

4. Output / Communication Protocol

Full spec: 3.二次开发/SDK(多种开发语言)/english/manual/UHF RFID Reader UHFReader18 User's Manual V2.0.doc. Same binary frame format is used regardless of transport (RS232 / RS485 / TCP).

Command frame (host → reader)

Len	Adr	Cmd	Data[]	CRC16-LSB	CRC16-MSB
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Response frame (reader → host)

Len	Adr	reCmd	Status	Data[]	CRC16-LSB	CRC16-MSB
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- Len = length of everything after itself (minimum 4 / 5)
- Adr = reader address; 0x00 default, 0xFF = broadcast
- CRC16: init 0xFFFF, poly 0x8408 (CCITT-reversed), LSB transmitted first

Key command codes

Inventory 0x01, Read Data 0x02, Write Data 0x03, Write EPC 0x04, Kill 0x05, Lock 0x06 ... plus reader-defined commands: Get Info 0x21, Set Region 0x22, Set Address 0x24, Set Baud 0x28, Set Power 0x2F, Set Wiegand 0x34, Set/Get WorkMode 0x35 / 0x36.

Real example capture (a tag read, active/EPC output)

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11 00 EE 00 E2 00 00 17 00 14 02 66 16 70 6B 48 83 37
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11 = Len, 00 = reader address, EE = active-output marker, 00 = status OK, remaining bytes = EPC ID + CRC16.

Output content is configurable via the “Set WorkMode” command (0x35): choose RS232/RS485 output vs. Wiegand output, and choose what is returned (Password / EPC / TID / User memory, or full EPC via the Mem_Inven byte). Wiegand format (26 vs. 34-bit, bit order, pulse timing) is set with command 0x34.

5. Writing Your Own Software

UHFRReader18.dll plus demo projects in C#, Delphi, VB6, Android, and a Web/ActiveX control are provided under 3.二次开发/SDK(多种开发语言)/english — all using the same protocol frames described above.

6. Support

From 9.技术支持/联系方式.txt:

- Contact: Engineer Guo (郭工) — phone/WeChat 15218866025, QQ 1525812579
- Hours: Monday–Saturday, 08:30–12:00 and 13:30–18:00
- Before contacting, confirm the exact model printed on the back label of your unit for faster support